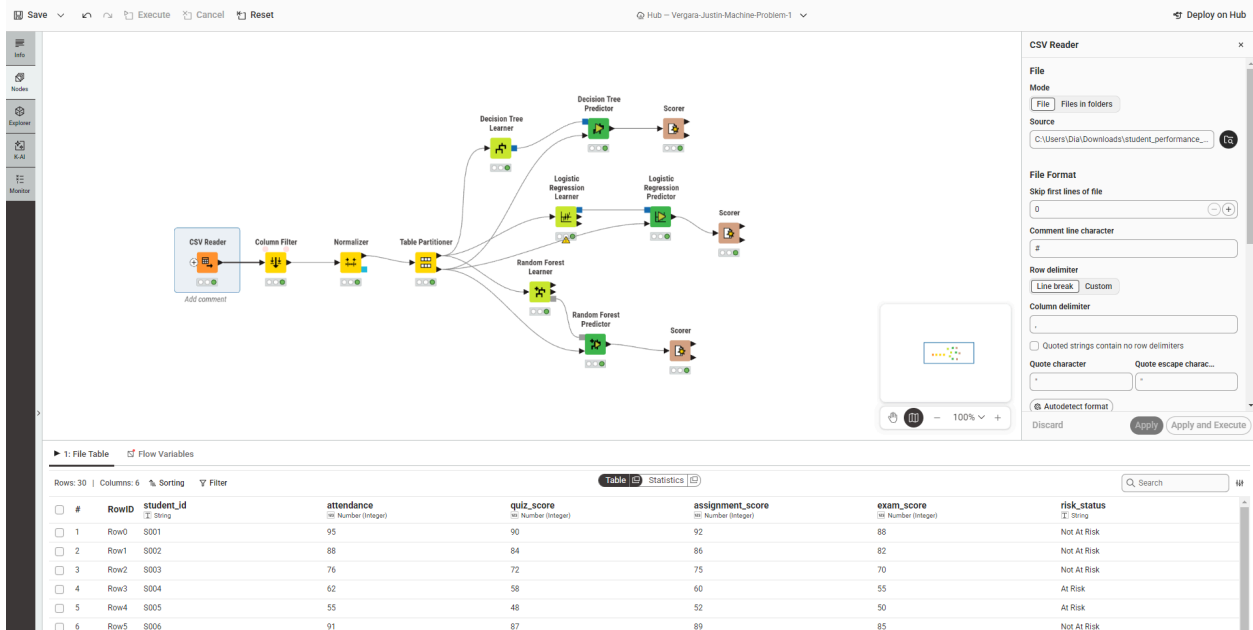


MACHINE PROBLEM 1 SCREENSHOTS

1. CSV READER (I borrowed Pardilla's Laptop because mine is still in the repair shop)



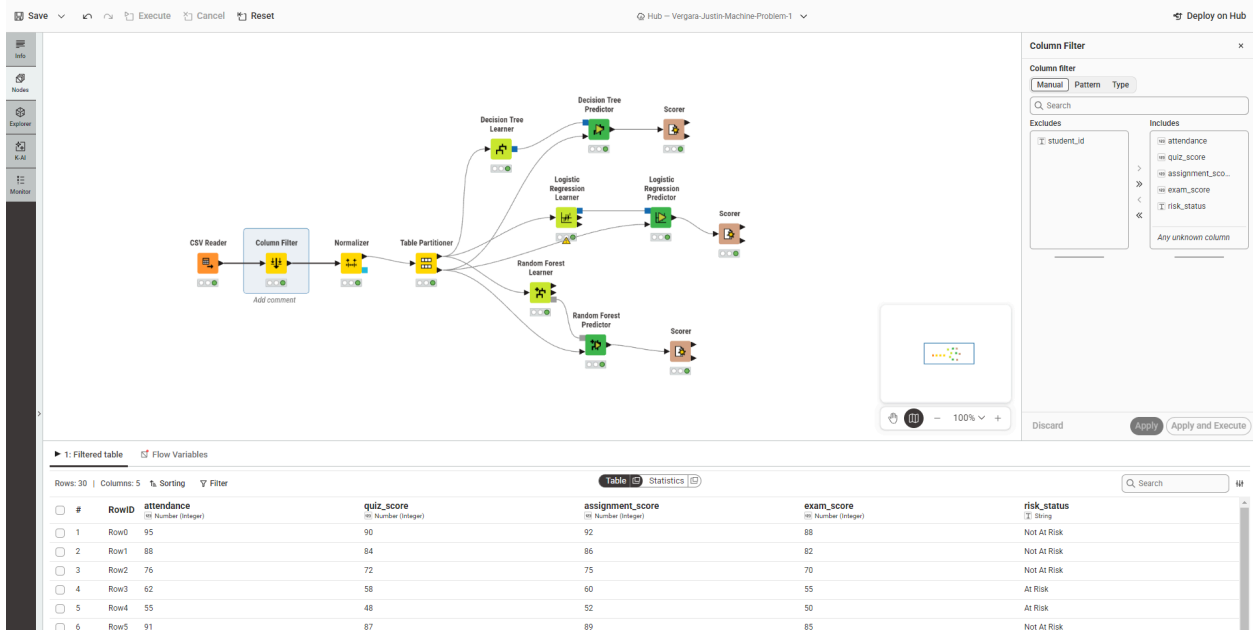
The screenshot shows the Orange3 Machine Learning interface. The workflow consists of the following nodes: CSV Reader, Column Filter, Normalizer, Table Partitioner, Decision Tree Learner, Decision Tree Predictor, Logistic Regression Learner, Logistic Regression Predictor, Random Forest Learner, Random Forest Predictor, and Scorer. The CSV Reader node is configured with the following settings:

- File: C:\Users\Jial\Downloads\student_performance...
- File Format: Skip first lines of file: 0
- Comment line character: #
- Row delimiter: Line break
- Column delimiter: Custom
- Quoted strings contain no row delimiters: ☐
- Quote character: "
- Quote escape character: \"
- Autodetect format: ☐

The data table below shows the first 6 rows of the dataset:

#	RowID	student_id	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	S001	95	90	92	88	Not At Risk
2	Row1	S002	88	84	86	82	Not At Risk
3	Row2	S003	76	72	75	70	Not At Risk
4	Row3	S004	62	58	60	55	At Risk
5	Row4	S005	55	48	52	50	At Risk
6	Row5	S006	91	87	89	85	Not At Risk

2. COLUMN FILTER



The screenshot shows the Orange3 Machine Learning interface with the Column Filter node selected. The workflow is the same as in the first screenshot. The Column Filter node is configured with the following settings:

- Column filter: Manual
- Excludes: student_id
- Includes: attendance, quiz_score, assignment_score, exam_score, risk_status

The data table below shows the first 6 rows of the filtered dataset:

#	RowID	attendance	quiz_score	assignment_score	exam_score	risk_status
1	Row0	95	90	92	88	Not At Risk
2	Row1	88	84	86	82	Not At Risk
3	Row2	76	72	75	70	Not At Risk
4	Row3	62	58	60	55	At Risk
5	Row4	55	48	52	50	At Risk
6	Row5	91	87	89	85	Not At Risk

3. NORMALIZER

4. TABLE PARTITIONER

The screenshot displays the Orange Data Mining environment. The main workspace contains a workflow diagram starting with a 'CSV Reader' node connected to a 'Column Filter'. This is followed by a 'Normalizer' node, which feeds into a 'Table Partitioner' node labeled 'Add comment'. From the 'Table Partitioner', five parallel paths emerge, each leading to a different model: 'Decision Tree Learner', 'Logistic Regression Learner', 'Random Forest Learner', 'Decision Tree Predictor', and 'Logistic Regression Predictor'. Each model's output is connected to a 'Scorer' node. Below the workflow, a small window shows a scatter plot of data points. On the right side, the 'Table Partitioner' configuration panel is open, showing settings for 'First partition type' (Relative (%)), 'Relative size' (80), 'Sampling strategy' (Random), 'Group column' ('risk_status'), and 'Fixed random seed' (1678807467440). At the bottom, a table titled 'Table Statistics' provides summary statistics for the dataset.

	#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status
☐	1	Row0	0.98	0.92	0.979	0.868	Not At Risk
☐	2	Row1	0.837	0.8	0.854	0.755	Not At Risk
☐	3	Row2	0.592	0.56	0.625	0.528	Not At Risk
☐	4	Row3	0.306	0.28	0.312	0.245	At Risk
☐	5	Row4	0.163	0.08	0.146	0.151	At Risk
☐	6	Row5	0.898	0.85	0.917	0.811	Not At Risk

5. DECISION TREE LEARNER

Save Execute Cancel Reset Q Hub - Vergara-Justin-Machine-Problem-1 Deploy on Hub

Info Nodes Explorer K.AI Monitor

1: Decision Tree Model Flow Variables

The data at the output port is not supported by any modern viewer.

Decision Tree Learner

Class column
risk_status

Quality measure
Gini index Gain ratio

Pruning method
No pruning MDL

☒ Reduced error pruning

Minimum number of records per node
2

Number of records to store for view
10000

Average split point

Number threads
12

☒ Skip nominal columns without domain information

☐ Force root split column

☐ Binary nominal splits

DAHLI Continue

Discard Apply Apply and Execute

6. DECISION TREE PREDICTOR

Save Execute Cancel Reset Q Hub - Vergara-Justin-Machine-Problem-1 Deploy on Hub

Info Nodes Explorer K.AI Monitor

1: Classified Data Flow Variables

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)	Prediction (risk_status) (# String)
☐	1	Row10 0.776	0.74	0.708	0.774	Not At Risk	Not At Risk
☐	2	Row12 0.653	0.6	0.667	0.585	Not At Risk	Not At Risk
☐	3	Row23 0.449	0.4	0.438	0.358	At Risk	At Risk
☐	4	Row24 1	1	1	1	Not At Risk	Not At Risk
☐	5	Row27 0.041	0	0.042	0	At Risk	At Risk
☐	6	Row29 0.51	0.44	0.479	0.396	At Risk	At Risk

Decision Tree Predictor

Number of patterns for hilling
10000

☐ Custom prediction column name

☐ Append columns with normalized class distribution

Discard Apply Apply and Execute

7. DECISION TREE SCORER

Save Execute Cancel Reset Q Hub - Vergara-Justin-Machine-Problem-1 Deploy on Hub

Info Nodes Explorer K.AI Monitor

1: Confusion matrix 2: Accuracy statistics Flow Variables

Rows: 2	Columns: 2	%	Sorting	Filter
#	RowID	Not At Risk	At Risk	
1	Not A...	3	0	
2	At Risk	0	3	

Table ID Statistics

Search

8. LOGISTIC REGRESSION LEARNER

Save Execute Cancel Reset Q Hub - Vergara-Justin-Machine-Problem-1 Deploy on Hub

Info Nodes Explorer K.AI Monitor

1: Model for Predictor 2: Coefficients and Statistics 3: Model and Learning Properties Flow Variables

Logistic Regression Learner

Target column: T: risk_status

Reference category: At Risk

Solver: Stochastic average gradient

Feature selection: Manual Pattern Type

Search

Excludes: No columns in this list.

Includes: attendance, quiz_score, assignment_s..., exam_score, Any unknown colu...

Use order from column domain

Discard Apply Apply and Execute

The data at the output port is not supported by any modern viewer.

9. LOGISTIC REGRESSION PREDICTOR

Save Execute Cancel Reset Hub - Vergara-Justin-Machine-Problem-1 Deploy on Hub

Info Nodes Explorer K-ML Monitor

Logistic Regression Predictor

- ☐ Custom prediction column name
- ☐ Append columns with normalized class distribution

Discard Apply Apply and Execute

► 1: Predicted data Flow Variables

Rows: 6 | Columns: 6 % Sorting Filter Table Statistics Search

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)	Prediction (risk_status) (# String)
☐	1	Row10 0.775	0.74	0.708	0.774	Not At Risk	Not At Risk
☐	2	Row12 0.653	0.6	0.667	0.585	Not At Risk	Not At Risk
☐	3	Row23 0.449	0.4	0.438	0.358	At Risk	At Risk
☐	4	Row24 1	1	1	1	Not At Risk	Not At Risk
☐	5	Row27 0.041	0	0.042	0	At Risk	At Risk
☐	6	Row10 0.51	0.54	0.475	0.564	At Risk	At Risk

10. LOGISTIC REGRESSION SCORER

Save Execute Cancel Reset Hub - Vergara-Justin-Machine-Problem-1 Deploy on Hub

Info Nodes Explorer K-ML Monitor

Scorer

First column
Prediction (risk_status)

Second column
risk_status

Sorting strategy
Insertion order

☐ Reverse order
☐ Use name prefix

Missing values
Ignore Fail

Discard Apply Apply and Execute

► 1: Confusion matrix ► 2: Accuracy statistics Flow Variables

Rows: 2 | Columns: 2 % Sorting Filter Table Statistics Search

#	RowID	Not At Risk (# Number (Integer))	At Risk (# Number (Integer))
☐	1	Not At Risk 3	0
☐	2	At Risk 0	3

11. RANDOM FOREST LEARNER

Save Execute Cancel Reset Hub - Vergara-Justin Machine Problem-1 Deploy on Hub

Info Nodes Explorer K-AI Monitor

Random Forest Learner

Attribute Selection

Target column: risk_status

Training attributes: Use column Use fingerprint

Attribute selection: Manual Pattern

Excludes: No columns in this list.

Includes: attendance, quiz_score, assignment_s..., exam_score

Tree Options

Split criterion: Discard Apply Apply and Execute

1: Out-of-bag Predictions 2: Attribute Statistics 3: Random Forest Model 4: Flow Variables

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)	P (risk_status=Not At Risk) (# Number (Float))	P (risk_status=At Risk) (# Number (Float))	risk_status (Out-of-bag) (# String)	risk_status (Out-of-ba... (# Number (Float))	model count (# Number (Integer))
1	Row0	0.98	0.92	0.979	0.868	Not At Risk					
2	Row1	0.837	0.8	0.854	0.755	Not At Risk					
3	Row2	0.592	0.56	0.625	0.528	Not At Risk					
4	Row3	0.306	0.28	0.312	0.245	At Risk					
5	Row4	0.163	0.08	0.146	0.151	At Risk					
6	Row5	0.868	0.84	0.917	0.811	Not At Risk					

12. RANDOM FOREST PREDICTOR

Save Execute Cancel Reset Hub - Vergara-Justin Machine Problem-1 Deploy on Hub

Info Nodes Explorer K-AI Monitor

Random Forest Predictor

Change prediction column name

Prediction column name: Prediction (risk_status)

Append overall prediction confidence

Append individual class probabilities

Suffix for probability columns

Use soft voting

1: Prediction output 4: Flow Variables

#	RowID	attendance (# Number (Float))	quiz_score (# Number (Float))	assignment_score (# Number (Float))	exam_score (# Number (Float))	risk_status (# String)	Prediction (risk_status) (# String)	Prediction (risk_status) (Confiden... (# Number (Float))
1	Row10	0.776	0.74	0.708	0.774	Not At Risk	Not At Risk	1
2	Row12	0.653	0.6	0.667	0.585	Not At Risk	Not At Risk	1
3	Row23	0.449	0.4	0.438	0.358	At Risk	At Risk	1
4	Row24	1	1	1	1	Not At Risk	Not At Risk	1
5	Row27	0.041	0	0.042	0	At Risk	At Risk	1
6	Row29	0.61	0.44	0.479	0.564	At Risk	At Risk	0.79

● RANDOM FOREST SCORER

Save

Execute

Cancel

Reset

Hub - Vergara-Justin-Machine-Problem-1

Deploy on Hub

Info

Nodes

Explorer

K-Alt

Monitor

```
graph LR; CSVReader[CSV Reader] --> ColumnFilter[Column Filter]; ColumnFilter --> Normalizer[Normalizer]; Normalizer --> TablePartitioner[Table Partitioner]; TablePartitioner --> DTL[Decision Tree Learner]; TablePartitioner --> LRL[Logistic Regression Learner]; TablePartitioner --> RFL[Random Forest Learner]; DTL --> DTP[Decision Tree Predictor]; LRL --> LRP[Logistic Regression Predictor]; RFL --> RFP[Random Forest Predictor]; DTP --> S1[Scorer]; LRP --> S2[Scorer]; RFP --> S3[Scorer]; RFP --> S4[Scorer];
```

Add comment

Scorer

First column
Prediction (risk_status)

Second column
risk_status

Sorting strategy
Insertion order

☐ Reverse order

☐ Use name prefix

Missing values
Ignore Fail

Discard Apply Apply and Execute

1: Confusion matrix

2: Accuracy statistics

Flow Variables

Rows: 2 | Columns: 2 | % Sorting | Filter

Table Statistics

Search

HI

#	RowID	Not At Risk (# Number (Integer))	At Risk (# Number (Integer))
1	Not At Risk	3	0
2	At Risk	0	3